

FeeSync Lite

Complete Fee Management for Solo Coaching Operators

PRODUCT FeeSync Lite — Single-Admin Edition

VERSION PRD v1.0 • April 2026

AUTHOR As • Solo Founder

STATUS Production-Ready Specification

Single Admin

Flutter + Web

All 6 Fee Plans

WhatsApp Native

9 AI Features

8 M+

Individual tutors
across India

0

Free tier up
to 50 students

< 3 min

Onboarding to
first payment

Flutter + Web

Both platforms
ship together

9

AI features
built in

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EXECUTIVE SUMMARY & VISION

Why FeeSync Lite Exists

The Indian private coaching market spans 1.5 million centers — but the vast majority are operated by a single individual who teaches, collects fees, manages parents, and tracks attendance all by themselves. FeeSync Lite is a cloud-native, mobile-first fee management platform purpose-built for these solo operators: home tutors, yoga instructors, music teachers, dance coaches, and single-room coaching owners. It digitises every financial and operational touchpoint — from student enrollment and batch assignment through fee collection, WhatsApp receipt dispatch, attendance tracking, and real-time P&L; visibility — with zero accounting knowledge required.

Value Proposition	Description
Zero Paper	100% digital fee records. Searchable by name, phone, or batch instantly.
WhatsApp-First	Instant receipt and reminder delivery via WhatsApp — the channel every Indian parent uses.
Offline-Tolerant	Flutter app caches data locally and syncs automatically when connectivity returns.
Full Fee Engine	All 6 fee plan types: monthly, quarterly, half-yearly, annual, custom installment, per-session.
Razorpay Native	UPI QR codes, payment links, online checkout, and auto-reconciliation built in.
AI-Powered	9 AI features: overdue predictor, expense OCR, anomaly detection, churn risk, smart summaries.
Zero Accounting Skills	Wizard-driven workflows. Any tutor can operate it from day one — no training needed.
Truly Affordable	■0 free tier for up to 50 students. ■199/month entry price for paid tier.

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MARKET CONTEXT & TARGET USERS

Personas & Pain Points

Primary Personas

Persona	Profile	Student Count	Primary Need
Solo Coaching Owner	Runs one center alone. Teaches 2–4 subjects. 10–150 40–150	10–150	Real-time revenue and overdue visibility
Home Tutor	Teaches from home. Class 8–12. Multiple subjects. 15–60 15–60	15–60	Fast cash/UPI recording and WhatsApp receipts
Yoga / Fitness Instructor	Morning and evening batches. Monthly membership 20–80 20–80 model.	20–80	Batch-wise monthly fee tracking
Music / Dance Teacher	Per-session or monthly. Attendance-linked billing. 10–50 10–50	10–50	Per-class payment recording and absence alerts
Spoken English / IELTS	Short 1–3 month intensive courses. Fixed course fees. 15–40 15–40	15–40	Course fee installment tracking
Art / Skill Coach	Weekend workshops. One-time or monthly fees. 8–30 8–30	8–30	One-time receipt and renewal reminder
Single-Subject Coaching	Physics / Maths / Chemistry specialist. One room. 50–150 50–150	50–150	Bulk overdue reminder and monthly P&L clarity

Pain Points Solved

Pain Point	Impact	FeeSync Lite Solution
Fee register or diary lost	All payment history gone permanently	Cloud-synced records — permanent, searchable, recoverable
No real-time overdue view	Owner only discovers defaults at month-end	Live dashboard — overdue count and amounts always visible

Pain Point	Impact	FeeSync Lite Solution
Manual receipt writing	Time-consuming, error-prone, leads to disputes	Auto-PDF receipt generated on payment, sent to WhatsApp in seconds
No bulk reminder tool	Admin calls each parent individually - Overhead	One-click bulk WhatsApp reminder to all overdue parents
Cash and UPI confusion	Cannot reconcile at month-end	Payment mode tracking with daily and monthly reconciliation report
No monthly P&L view	Owner never knows true profitability	Automated P&L: revenue minus expenses per month, exportable PDF
Student data lost on exit	No history if student leaves or disputes arise	Soft-delete — all history preserved permanently
Multiple fee arrangements	Different deals per student, impossible to track	Per-student fee assignment with discount and waiver support
No attendance alerts	Parents unaware when child misses class	Auto WhatsApp absent alert sent to parent the same day

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PRODUCT ARCHITECTURE

Stack, Deployment, Database Schema

Technology Stack

Layer	Technology	Purpose / Notes
Frontend (Web)	React 18 + Vite + TypeScript	Fast development with HMR, type safety, Vercel edge deployment
Styling	Tailwind CSS + shadcn/ui	Consistent, accessible, mobile-first UI components
State Management	Zustand + React Query	Lightweight global state and server-state synchronisation
Backend / DB	Supabase — PostgreSQL + Auth + Realtime	Fully-managed backend with Row-Level Security and instant APIs
File Storage	Supabase Storage	Receipts, student documents, expense photos, logos
Deployment (Web)	Vercel (edge functions)	Zero-config CI/CD, global CDN, environment-based builds
Mobile App	Flutter 3 + Dart	Single codebase for Android and iOS — com.feesync.lite
Payments	Razorpay Payment Gateway	UPI, cards, netbanking, QR codes, webhook reconciliation
WhatsApp	Meta Cloud API / Interakt / WATI	Pre-approved HSM templates — receipts and all 14 notification types
SMS Fallback	Fast2SMS / MSG91	Automatic fallback when WhatsApp delivery fails
Email	Resend.com	Transactional emails — receipts, reports, password reset
PDF Generation	React-PDF (client) + Puppeteer (edge)	Receipts generated on device; reports generated server-side
AI Features	Gemini API / OpenAI GPT-4	Powers all 9 AI features — overdue predictor, OCR, NLQ, summaries
Error Monitoring	Sentry	Production error tracking and alerting
Analytics	Supabase PostgREST + custom views	Internal dashboards and KPI computation — no extra cost

Database Schema — Core Tables

Architecture: FeeSync Lite uses a single-admin model. Each coaching center has one *center_profile* record and one *owner_user* account. All tables are scoped to **owner_id** via Row-Level Security — the owner can only ever see their own data. There are no branch-level scopes, no role hierarchies, and no cross-tenant queries possible.

Table	Purpose	Key Fields
center_profile	Single center record. One row per owner.	owner_id, center_name, logo_url, address, gstin, receipt_prefix, default
students	Student profiles. Linked to center and batch.	owner_id, name, parent_phone, whatsapp_no, batch_id, status, admissi
batches	Subject or batch configuration.	owner_id, name, subject, teacher_name, capacity, fee_amount, schedu
fee_plans	Structured fee plans of all 6 types.	owner_id, type, amount, due_day, late_fine, grace_days, gst_percent
fee_assignments	Student ↔ fee plan mapping with discount.	student_id, fee_plan_id, start_date, discount_amount, waiver_reason
payments	Every payment transaction.	student_id, amount, mode, receipt_no, utr_ref, created_at
dues	Auto-computed outstanding per student per period.	Materialised view — student_id, period, assigned, paid, outstanding

Table	Purpose	Key Fields
expenses	Center expense entries.	owner_id, category, amount, vendor, payment_mode, receipt_url, date
attendance	Daily attendance record per student per batch.	student_id, batch_id, date, status (present/absent/late)
notifications_log	Delivery log for all sent WhatsApp, SMS, email messages.	student_id, type, channel, status, sent_at
subscriptions	SaaS plan and billing state for the center.	owner_id, plan_tier, valid_until, max_students, razorpay_sub_id

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AUTHENTICATION & ONBOARDING

Login, Session, Setup Flow

FeeSync Lite uses a single-admin model. There is exactly one account per center. The owner is the only user. There are no staff invites, no permission tables, and no role-based access control. Authentication is handled by Supabase Auth. The onboarding flow is designed to reach the first useful action — adding a student and recording a payment — in under three minutes.

Onboarding Flow

Step	Screen	Required Input	Time Target
1	Landing → Register	Email and password, or Google OAuth sign-in	30 s
2	Center Setup	Center name, what you teach (e.g. Maths, Yoga), default monthly fee	60 s
3	Profile (optional)	Logo, address, GSTIN — all skippable, editable later in Settings	0–45 s
4	Dashboard (empty state)	'Add your first student' CTA displayed prominently with progress strip	
5	Add First Student	Student name, parent phone number, batch, fee plan	45 s
6	First Payment (guided)	Optional walkthrough: record a payment and preview the WhatsApp receipt	60 s

Authentication Specifications

Auth Aspect	Specification
Login Method	Email + password or Google OAuth. Phone OTP login planned for v2.
Session	Short-lived JWT plus refresh token. Auto-expire after 30 days of inactivity.
Two-Factor Auth	TOTP-based 2FA — prompted at first login, skippable once. Mandatory for account security.
Password Reset	Email reset link via Supabase Auth. Standard secure flow.
Account Deletion	Owner can self-delete the account. All data erased within 30 days per DPDP Act 2023.

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DASHBOARD & ANALYTICS

Owner Control Centre

The dashboard is the owner's primary control centre. It answers three questions at a glance: how much did I collect this month, who has not paid, and what do I need to do right now. It combines live KPI cards, trend charts, and a quick action bar — all on a single screen.

Dashboard KPI Cards — Always Visible

KPI Card	Data Logic	Priority
Total Collection Today	Sum of all payments recorded today	CORE
This Month Revenue	Sum of all payments in the current calendar month	CORE
Total Overdue Amount	Sum of all outstanding dues across every active student	CORE
Overdue Students Count	Count of students with any outstanding due amount	CORE

KPI Card	Data Logic	Priority
Active Students	Count of students with status = active	CORE
New Admissions This Month	Count of students enrolled in the current month	HIGH
Attendance Today %	Present count divided by total enrolled across all batches	HIGH
Upcoming Renewals (7 days)	Students with fee due in the next 7 days	HIGH

Dashboard Charts

- Monthly Revenue Bar Chart — last 12 months, colour-coded by payment mode (cash, UPI, cheque)
- Collection vs Due Line Chart — daily totals for the current month, highlights the overdue gap
- Student Growth Line — enrollment count trend by month
- Batch Strength Donut — students per batch with capacity utilisation percentage
- Payment Mode Pie — cash vs UPI vs cheque share for the current month
- Top Defaulters Widget — top 10 overdue students with amount owed and days overdue
- Today's Collection Activity Feed — live stream of payments recorded today
- Upcoming Due Dates Timeline — 14-day calendar strip with amounts due each day

Quick Action Bar

- + Record Payment
- + Add Student
- Send Bulk Reminders
- Download Monthly Report
- Mark Attendance (Today)
- View Overdue List

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STUDENT MANAGEMENT

Profiles, Batches, Documents, Lifecycle

Student Profile Fields

Personal Information	Academic Information
Full Name (required)	Class / Standard
Date of Birth	School or College Name
Gender	Aadhaar Number (optional, stored masked)
Photo (optional)	Enrollment Date (auto-filled)
Primary Phone (student)	Admission Number (auto-generated)
Parent / Guardian Name	Batch / Subject (multi-batch supported)
Parent Phone (required — WhatsApp)	Status: Active / Inactive / Alumni
WhatsApp Number	Notes / Remarks
Email (optional)	Discount or Waiver flag
Address	—

Student Actions

Action	Description	Priority
Enroll in Batch	Add student to one or more batches. Each enrollment triggers a fee assignment.	CORE
Record Payment	Quick payment from student profile. Pre-fills student and fee plan details.	CORE
View Fee History	Full payment timeline with receipt links. Filter by month or year.	CORE
View Dues	Outstanding summary — per-plan breakdown with days overdue.	CORE
Send WhatsApp	Manual WhatsApp message to parent. Pre-built templates available.	CORE
Upload Document	Aadhaar, birth certificate, or photo ID. Stored in Supabase Storage.	HIGH
Apply Discount / Waiver	One-time or recurring discount. Reason required and logged.	HIGH
Deactivate / Graduate	Soft-delete. Student becomes alumni — all history permanently preserved.	CORE
Merge Duplicate	Merge two accidentally created duplicate records. Payments auto-merged.	HIGH
Print ID Card	One-click ID card PDF with photo, name, batch, and admission number.	MEDIUM
Bulk Import via CSV	Import hundreds of students from Excel or CSV with column mapping.	HIGH
Add Note	Free-text internal note on any student — visible only to the admin.	HIGH

Student List Features

- Global search by name, phone number, parent phone, or admission number
- Filter by batch, status (active/inactive/alumni), overdue flag, and enrollment month

- Sort by name, dues amount, or last payment date — overdue-first sort available
- Bulk select rows: send WhatsApp, export selected, or change status in bulk
- Colour-coded rows: red = overdue more than 30 days, orange = due this week, green = paid up

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FEE MANAGEMENT CORE

Plans, Collection, Receipts, Dues Engine

Fee Plan Types

Plan Type	Description	Priority
Monthly Fixed	Fixed amount per month. Auto-generates a due record on a configurable day each month.	CORE
Quarterly	3-month bundle. Pro-rated automatically if student enrolls mid-quarter.	CORE
Half-Yearly	6-month plan with full pro-rata support for mid-cycle enrollments.	HIGH
Annual / One-Time	Single fixed amount for the full year or a one-time enrollment fee.	CORE
Custom / Installment	Any amount on any date. Example: ₹5,000 on 15th January, ₹5,000 on 15th February.	HIGH
Per-Class / Session	Amount charged per class attended. Linked directly to attendance records.	CORE

Fee Plan Configuration

- Base amount (required)
- Late fine per day after due date (optional) with configurable grace period in days
- Tax rate: GST percentage — 0%, 5%, 12%, or 18% — for registered centers
- Auto-generate monthly due records: toggle on or off per plan
- Discount categories: sibling discount, merit discount, staff-child discount
- Plan name, description, and applicable batches

Payment Recording — 7-Step Flow

Step	Action	Details
1	Search student	Type name or phone number — instant autocomplete from all active students.
2	Select fee period	System shows all pending dues. Select which period or periods to pay.
3	Enter amount	Pre-filled with the due amount. Partial payment supported — tracked separately.
4	Select payment mode	Cash / UPI / Cheque / Bank Transfer / Card / Scholarship.
5	Enter reference	UPI UTR number, cheque number, or transaction ID — required for non-cash modes.
6	Add discount / waiver	Optional. Reason required and saved to history.
7	Confirm and receipt	Receipt number auto-assigned. PDF generated instantly. WhatsApp receipt sent.

Receipt Design

- Center logo, name, address, and GSTIN
- Auto-assigned receipt number in configurable format (e.g. FS-2024-0001)
- Student name, admission number, and batch name
- Fee period description (e.g. 'December 2024 — Physics Batch A')
- Amount paid, payment mode, and reference number

- Outstanding balance remaining after this payment
- Optional digital signature field
- QR code linking to the online receipt verification page
- Printable A5 format and WhatsApp-optimised A6 format
- DUPLICATE watermark applied automatically on reprints

Dues Engine

The dues engine auto-generates pending fee records at the start of each billing period. A Supabase Edge Function runs as a CRON job daily at midnight IST. It re-computes all pending dues, accounts for enrollment date pro-rating, applies discounts and late fines that accumulate daily after the grace period, handles partial payment tracking, and processes mid-cycle plan changes. Newly overdue students are flagged automatically and queued for WhatsApp reminder dispatch.

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PAYMENT INTEGRATION

Razorpay, UPI QR, Reconciliation

Supported Payment Modes

Mode	Description	Priority
Cash	Recorded manually by admin. No payment gateway required. Most common mode.	CORE
UPI (manual entry)	Admin enters the UTR number from the parent's UPI transfer. Manual verification.	CORE
Razorpay Payment Link	Admin generates a payment link and sends to parent via WhatsApp. Parent pays online.	HIGH
Razorpay UPI QR	Static QR code per center and dynamic QR code per invoice. Parent scans and pays.	HIGH
Razorpay Subscription	Auto-debit monthly with parent consent. Managed via Razorpay subscription API.	MEDIUM
Cheque	Cheque number, bank name, and date recorded. Status: received / cleared / bounced.	CORE
Bank Transfer / NEFT	Reference number logged. Manual verification by admin.	CORE
Scholarship / Waiver	Zero-amount transaction. Reason required and stored in history.	CORE

Online Payment Flow (Razorpay)

- Admin generates a payment link from the student's pending due record
- Link is sent to the parent via WhatsApp, SMS, or email
- Parent opens link and completes payment through Razorpay checkout (UPI, card, or netbanking)
- Razorpay webhook fires to the Vercel Edge Function
- Edge Function verifies the webhook signature and records the payment in Supabase
- PDF receipt is generated automatically and WhatsApp receipt is sent to the parent
- Dashboard KPI cards update in real time via Supabase Realtime subscription

Cheque Lifecycle

Cheques move through a defined lifecycle: Received → Deposited → Cleared or Bounced. On a bounce event, the payment is automatically reversed, the student's outstanding due is restored, a configurable bounce charge is added, and an alert is sent to the admin. A clearance reminder is triggered three days after the deposit date if the status remains Deposited.

Reconciliation Report

The daily and monthly reconciliation view shows total collected by payment mode, Razorpay settlement amounts versus manually recorded entries, pending cheque clearances, and any discrepancies flagged in red. Exportable as Excel.

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NOTIFICATIONS & COMMUNICATION

WhatsApp, SMS, In-App, Email

Notification Triggers

Notification	Trigger	Channel	Priority
Fee Receipt	Immediately on payment received	WhatsApp + Email	CORE
Due Reminder — 3 days before	Automated CRON	WhatsApp	CORE
Due Reminder — On due date	Automated CRON	WhatsApp + SMS	CORE
Overdue Alert — 7 days	Automated CRON	WhatsApp	CORE
Overdue Alert — 15 and 30 days	Automated CRON	WhatsApp + SMS	HIGH
Payment Link	Manual by admin	WhatsApp	HIGH
Admission Confirmation	On student creation	WhatsApp + Email	CORE
Batch Schedule Change	Manual by admin	WhatsApp	HIGH
Holiday / Closure	Manual broadcast	WhatsApp	HIGH
Fee Plan Change	On plan update	WhatsApp	HIGH
Attendance Absent Alert	Daily, if enabled per batch	WhatsApp to parent	HIGH
Exam / Test Reminder	Manual by admin	WhatsApp	MEDIUM
Center-level Broadcast	Manual by admin	WhatsApp + SMS + In-app	HIGH
Subscription Renewal Reminder	Auto, 7 days before expiry	Email + In-app	CORE

WhatsApp Template Management

- All outgoing messages use pre-approved Meta HSM (Highly Structured Message) templates
- Template variables: {{student_name}}, {{amount}}, {{due_date}}, {{center_name}}, {{receipt_no}}
- Admin can customise the template body text within the variable constraint boundaries
- Language options: English, Hindi, Bengali, Assamese, Tamil, Telugu — selectable per center
- Parent opt-out management: if a parent opts out, the system respects it automatically
- Delivery status tracked per message: sent / delivered / read / failed
- Failed WhatsApp messages automatically fall back to SMS via the configured SMS provider

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REPORTS & EXPORTS

14 Report Types — Full Power

Report	Description	Export	Priority
Daily Collection Report	All payments today. Grouped by payment mode.	PDF / Excel / CSV	CORE
Monthly Revenue Report	Total collected, total due, total outstanding. Month-over-month comparison.	PDF / Excel	CORE
Overdue / Due Report	All students with outstanding dues. Sortable by amount and days overdue.	PDF / WhatsApp	CORE
Fee Plan Report	Revenue per fee plan — enrolled count, collected, outstanding.	PDF	HIGH
Student Ledger	Complete financial history for one student. Printable.	PDF / WhatsApp	CORE
P&L Statement	Revenue minus expenses. Monthly and annual view. Accrual and cash basis.	PDF / Excel	HIGH
Expense Report	All expenses by category, vendor, and date range	PDF / Excel	HIGH
Attendance Report	Student-wise attendance percentage for any date range	PDF / Excel	HIGH
Defaulter List	Students overdue beyond a configurable number of days	PDF / WhatsApp	CORE
Collection Efficiency	Target versus actual collection percentage per month	PDF	HIGH
New Admissions Report	New enrollments per month and batch.	PDF	MEDIUM
Alumni / Dropout Report	Students who left, reason for leaving, and revenue lost.	PDF	MEDIUM
GST Summary	GST collected by tax slab. Formatted for accountant.	PDF / Excel	HIGH
Payment Mode Analysis	Cash, UPI, and cheque trends and share over time	PDF	MEDIUM

Export Formats

- PDF — print-ready, branded with center logo and name
- Excel (.xlsx) — all calculated columns preserved, suitable for an accountant
- CSV — raw data dump for use in external tools or custom analysis
- WhatsApp — share any report as a PDF link directly via WhatsApp

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ATTENDANCE MODULE

Daily Marking, Reports, Parent Alerts

Feature	Description	Priority
Batch-wise daily attendance	Admin selects a batch and date. Sees the full roster. Marks Present, Absent, or Late.	CORE
Bulk present (default all present)	One-click to mark everyone present, then individually flag absences. Faster for large batches.	CORE
Attendance backdating	Mark attendance for up to 7 days in the past. Reason required. Action is logged.	HIGH
Absent parent alert	WhatsApp alert sent to parent automatically when child is marked absent. Configurable toggle.	HIGH
Attendance % on student profile	Running attendance percentage. Alert displayed when student falls below 75%.	HIGH
Attendance register PDF	Monthly attendance register for a batch. Printable A4 landscape format.	HIGH
Holiday management	Mark center holidays. Attendance is automatically excluded for those dates.	HIGH

Feature	Description	Priority
Per-session fee billing link	Attendance records drive the billing amount for per-session fee plans.	CORE
Leave request (v2)	Student submits a leave request via the parent portal. Admin reviews and approves.	MEDIUM

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BATCH & SCHEDULE MANAGEMENT

Timetable, Capacity, Clash Detection

Batch Configuration

- Batch name (e.g. 'Class 10 Science Morning')
- Subject or course name
- Teacher name — free text field, useful for logging freelance teacher names
- Room or location (optional)
- Capacity — maximum number of students, with 80% and 100% utilisation alerts
- Schedule — days of week plus time slots (supports multiple slots per day)
- Fee amount linked to a specific fee plan
- Start date and end date for the academic session
- Status: active, completed, or upcoming

Batch Operations

Operation	Description	Priority
Transfer students between batches	Move students with or without a fee plan change.	CORE
Clone batch for new session	Copy schedule, fee plan, and teacher name — starts with an empty enrollment	HIGH
Capacity alerts	Warning when batch exceeds 80% capacity. New enrollment blocked at 100%	HIGH
Batch-wise WhatsApp broadcast	Send a WhatsApp message to all parents of a specific batch at once.	HIGH
Weekly timetable grid view	Visual timetable grid. Clash detection: blocks two batches sharing the same time slot.	HIGH
Printable timetable PDF	Formatted timetable PDF for display at the center.	MEDIUM
Archive completed batch	Freezes all records. Students move to alumni status or transfer to a new batch.	CORE

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EXPENSE TRACKER & P&L

Center Costs, Profit & Loss Statement

Expense Categories

Rent	Teaching Material
Salaries and Freelance Fees	Equipment Purchase
Electricity and Utilities	Cleaning and Maintenance
Internet and Phone	Software and Subscriptions
Stationery and Printing	Transport
Marketing and Advertising	Miscellaneous

Expense Entry Fields

- Date, category (from dropdown — customisable), amount

- Vendor or payee name
- Payment mode: cash, UPI, or bank transfer
- Reference number (for digital payments)
- Receipt photo upload — stored in Supabase Storage
- Notes field for additional context
- Recurring flag — for monthly expenses like rent that repeat automatically

P&L; Statement

The Profit & Loss statement is generated automatically from fee collection data and expense entries. Revenue equals total fee payments received. Expenses are drawn from all expense entries in the selected period. Net Profit equals Revenue minus Expenses. The statement is displayed as a monthly bar chart and a detailed table. Both accrual basis and cash basis views are available. Annual view is supported. Exportable as a PDF formatted for a CA or accountant.

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PARENT PORTAL — FLUTTER APP

com.feesync.lite — Android + iOS

The Flutter parent app ships alongside the web app — it is not a future milestone. Package ID: **com.feesync.lite**. The web admin panel is also a fully responsive Progressive Web App installable on Android via 'Add to Home Screen'.

Parent App Screens

Screen	Description	Priority
Login	Phone number OTP login. Linked automatically to the parent phone in the student record.	CORE
Home	Child's name, current month fee status (paid, due, or overdue), and next due date.	CORE
Fee History	Complete payment history. Tap any record to view or download the receipt PDF.	CORE
Current Dues	Outstanding dues per period. 'Pay Now' button initiates Razorpay checkout.	CORE
Attendance	Child's running attendance percentage and a daily calendar (present/absent).	HIGH
Timetable	Child's batch schedule for the current week.	HIGH
Notifications	In-app notification feed — receipts, due reminders, and center announcements.	CORE
Contact Center	Phone, WhatsApp, and email buttons to reach the admin directly.	HIGH
Multiple Children	Switch between children if more than one child is enrolled at the same center.	HIGH
Dark Mode	Full dark mode support throughout the app.	MEDIUM
Offline Support	Last-known data is cached locally. App remains usable without internet connectivity.	HIGH

Flutter Technical Requirements

Aspect	Specification
Framework	Flutter 3.x with Dart null-safety enforced throughout
Backend	Supabase Flutter SDK for authentication and data access
State Management	Riverpod for scalable and testable state management
Local Cache	Hive for offline data storage and sync queue
Payments	Razorpay Flutter plugin for in-app checkout
Push Notifications	firebase_messaging for remote push notifications
In-App Alerts	flutter_local_notifications for local notification display
Receipt Viewer	pdf and printing packages for in-app PDF view and download
Package ID	com.feesync.lite
Android Support	Min SDK: 21 (Android 5.0) — Target SDK: 34 (Android 14)
Security	ProGuard / R8 obfuscation enabled in release builds

Aspect	Specification
Distribution	Signed release APK and App Bundle for Google Play Store submission

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AI-POWERED FEATURES

9 Smart Features — Built In

FeeSync Lite ships with nine AI-powered features designed to reduce manual work, surface actionable insights, and help the admin stay ahead of collection and student retention problems — without requiring any AI expertise to use.

AI Feature	Description	Priority
Smart Overdue Predictor	Machine learning model trained on payment history. Flags students likely to default before the due date, enabling proactive outreach.	HIGH
Natural Language Fee Query	Admin types a question like 'Who has not paid this month?' and receives an instant answer. Powered by Generative AI.	MEDIUM
AI Receipt OCR	Upload a photo of a manual or paper receipt. AI auto-extracts the amount, date, and payment mode. Useful for reconciling cash payments.	HIGH
Auto-categorise Expenses	Upload a photo of an expense bill. AI extracts the amount and vendor and suggests the appropriate expense category.	MEDIUM
WhatsApp Tone Optimizer	AI suggests more effective and polite fee reminder phrasing based on the parent's payment history and response rate.	LOW
Enrollment Growth Forecast	Analyses seasonal enrollment trends and predicts expected new admissions for the following month.	MEDIUM
Payment Anomaly Detection	Flags unusual payment patterns — large unexplained cash amounts, same-day cancel-and-re-record actions, or payments from unknown sources.	HIGH
Smart Daily Summary Report	AI-generated daily summary: 'Today you collected ₹X from Y students. 3 students are 30+ days overdue.'	MEDIUM
Student Churn Risk Score	Students with declining attendance and unpaid dues receive a high churn risk score. Admin is alerted proactively.	MEDIUM

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SETTINGS & CONFIGURATION

Center Profile, Fees, Notifications, Security

Settings are organised into six sections. All fields are optional at setup and can be updated at any time. The admin accesses Settings from the bottom navigation bar.

Center Profile

- Center name, logo, and tagline
- Address, phone number, email, and website
- GSTIN and PAN number (for invoices and receipts)
- Academic session start and end dates (e.g. April 2024 – March 2025)
- UPI ID — printed on receipts for parent reference

Fee Settings

- Receipt number format: configurable prefix and starting number
- Default due day of month (default: 1st)
- Late fine per day and grace period in days
- GST rate: 0%, 5%, 12%, or 18%
- Enable or disable online payments via Razorpay
- Discount categories and maximum percentage allowed per category

Notifications

- Enable or disable each notification type individually
- WhatsApp provider selection: Interakt, WATI, or Meta Direct API
- SMS provider selection: Fast2SMS or MSG91
- Reminder timing: number of days before due date (configurable)
- Absent alert toggle and delay in minutes after class time

Integrations

- Razorpay API key and secret
- WhatsApp Business API credentials
- Google Calendar sync for batch schedules
- Webhook URL for custom integrations (v2)

Appearance

- Primary brand colour via hex colour picker
- Dark or light mode default
- Receipt logo position and size
- Dashboard layout preference

Security

- TOTP 2FA — mandatory at first login, skippable once

- Session timeout duration
- On-demand full data export (CSV and JSON)
- Delete center account with full data erasure within 30 days (DPDP / GDPR)

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SUBSCRIPTION & BILLING

Right-Sized Pricing for Solo Operators

Pricing is calibrated so that a tutor with 50 students pays less per month than what they earn from a single student session. The free tier is genuinely useful — not a crippled demo. The paid tier unlocks full power at an accessible price.

Feature	Free	Starter ■199/mo	Growth ■499/mo
Max Active Students	50	200	Unlimited
All 6 Fee Plan Types	✓	✓	✓
WhatsApp Receipts	200 / month	Unlimited	Unlimited
WhatsApp Auto-Reminders	50 / month	Unlimited	Unlimited
SMS Fallback	—	100 / month	500 / month
Razorpay Online Payments	—	✓	✓
Razorpay Auto-Debit	—	—	✓
Reports	5 basic	All 14	All 14 + scheduled email
AI Features	—	3 features	All 9 features
Flutter Parent App	✓	✓	✓
Expense Tracker & P&L	Basic	Full	Full
Attendance Module	✓	✓	✓
CSV Data Export	—	✓	✓
WhatsApp Support	—	—	✓

Paywall & Business Rules

- Plan limits are enforced at the API level via Supabase RPC — they cannot be bypassed through the UI
- When the student limit is reached: a friendly upgrade card appears on the dashboard — not a blocking modal
- Expired subscription: read-only mode for 7 days, then restricted to the dashboard view only
- Annual plan: 2 months free — displayed as a rupee saving amount, not as a percentage
- All new signups receive 30 days of full Growth tier features — no credit card required
- Referral program: ■100 off the first month for both the referrer and the new signup

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SECURITY & COMPLIANCE

DPDP Act, RLS, 2FA, Data Protection

Control	Description	Priority
Row-Level Security (RLS)	Every table is scoped to owner_id. No cross-account data access is architecturally possible.	CORE
HTTPS / TLS 1.3	Vercel enforces TLS everywhere. No unencrypted HTTP connections permitted.	CORE
Supabase Auth (JWT)	Short-lived JWT tokens with refresh. Sessions are revoked immediately on account deletion.	CORE
Input Sanitisation	All user input is sanitised. Parameterised queries throughout. No raw SQL from the client.	CORE
File Upload Validation	MIME type and size validation on every upload. JPEG and PDF only. Private storage bucket.	CORE
TOTP Two-Factor Auth	TOTP-based 2FA mandatory prompt at first login. Admin may skip once but is reminded.	HIGH
Rate Limiting	Vercel Edge rate limiting on all authentication endpoints. Anti-brute-force protection.	HIGH
PII Masking	Aadhaar numbers are stored as a hashed value. Displayed as XXXX-XXXX-1234 only.	HIGH
Data Residency	Supabase region: ap-south-1 (Mumbai). All data remains within India at all times.	HIGH
Soft Deletes	Records are never hard-deleted. A deleted_at timestamp is set. Data is fully recoverable.	CORE
DPDP Act 2023 Compliance	WhatsApp consent collected at enrollment. Data erasure within 30 days. Full data export on request.	HIGH

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NON-FUNCTIONAL REQUIREMENTS

Performance, Scale, Offline, Accessibility

Requirement	Target	How Achieved
Page Load (web, LCP)	< 1.5 s on 4G	Vercel Edge CDN, Vite code splitting, lazy component loading
API Response (p95)	< 300 ms	Supabase connection pooler, indexed queries, materialised views
Mobile App Startup	< 2 s cold start	Flutter AOT compilation, optimised splash screen
Offline Mode (Flutter)	Full read + partial write	Hive local cache with sync queue on reconnect
Offline Mode (Web)	Full read + draft payment	Workbox Service Worker with IndexedDB persistence
Uptime SLA	99.9%	Vercel and Supabase managed infrastructure with PITR
PDF Receipt Generation	< 3 s	React-PDF client-side rendering, Puppeteer edge fallback
WhatsApp Delivery	< 10 s from payment record	Async queue via Supabase Edge Function
Database Scale	1M+ rows per table	PostgreSQL with owner_id-indexed queries throughout
Mobile Screen Support	360 px – 480 px optimised	Tailwind mobile-first breakpoints applied throughout
File Upload Limit	Max 10 MB per file	Supabase storage policy enforced and client-side validation
Backup	Daily automated	Supabase PITR (point-in-time recovery)
Browser Support	Chrome 90+, Firefox 88+, Safari 14+, and all modern CSS and JS, no IE support	
Accessibility	WCAG 2.1 AA	Semantic HTML, ARIA labels, 4.5:1 minimum contrast ratio

Requirement	Target	How Achieved
Internationalisation	English now. Hindi in v2. Bengali and Assamese in v3	Initial design and development from MVP

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RELEASE ROADMAP

MVP → v1 → v2 → v3

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MVP — MONTH 1–2

• Student management: add, edit, deactivate, bulk CSV import	CORE
• Fee plans: Monthly Fixed, One-Time, and Custom Installment	CORE
• Payment recording: Cash, UPI, Cheque	CORE
• PDF receipt generation — client-side via React-PDF	CORE
• Auto WhatsApp receipt on every payment record	CORE
• Dues engine CRON — Supabase Edge Function, daily at midnight IST	CORE
• Dashboard: 6 KPI cards, overdue widget, collection activity feed	CORE
• Basic daily collection report and monthly revenue report	CORE
• Email + password auth and Google OAuth — single admin login	CORE
• Subscription billing: Free tier (50 students) + Starter ■199/mo	CORE
• Flutter parent app MVP: Login, Home, Fee History, Current Dues	CORE

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V1.0 — MONTH 3–4 • FULL FEATURE PARITY

• Per-session, quarterly, and half-yearly fee plans	CORE
• Automated WhatsApp reminders: 3 days before, due date, 7 days overdue, 15 and 30 days	CORE
• Bulk 'Remind All Overdue' — one tap sends to all overdue parents	CORE
• Attendance module: daily marking, parent absent alert, attendance register PDF	CORE
• Expense tracker and automated P&L; statement	HIGH
• Razorpay payment link and UPI QR code integration	HIGH
• All 14 report types with PDF, Excel, and CSV export	HIGH
• Batch and timetable management with time-slot clash detection	HIGH
• Full TOTP 2FA implementation	HIGH
• Flutter parent app: Attendance, Timetable, Notifications, Razorpay Pay Now	HIGH
• Growth tier ■499/mo with AI features toggle	HIGH
• WhatsApp template management and language selector	HIGH
• Student ID card PDF generator	MEDIUM

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V2.0 — MONTH 5–7 • AI + SCALE

• Smart Overdue Predictor — ML on payment patterns	HIGH
• AI Receipt OCR — migrate paper records by photo	HIGH
• Payment Anomaly Detection — unusual pattern flagging	HIGH
• Student Churn Risk Score — dashboard widget	MEDIUM
• Daily AI Smart Summary Report	MEDIUM
• Auto-categorise expenses from receipt photo	MEDIUM
• Natural Language Fee Query via Gemini integration	MEDIUM

• Hindi UI language toggle throughout the web app	HIGH
• Bengali and Assamese WhatsApp template language support	HIGH
• Razorpay subscription auto-debit for Growth tier	MEDIUM
• Scheduled report emails: weekly and monthly to admin	HIGH
• iOS Flutter parent app — App Store submission	MEDIUM
• Duplicate student merge tool	HIGH

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V3.0 — MONTH 8–12 · POWER FEATURES

• Advanced analytics: cohort retention, LTV, churn curve charts	HIGH
• Offline-first web app: full PWA with background sync	HIGH
• Exam and test scheduling module	MEDIUM
• Course completion certificate generator	MEDIUM
• Enrollment Growth Forecast — seasonal AI prediction	MEDIUM
• Tally export format for accountants and CAs	MEDIUM
• WhatsApp Message Tone Optimizer AI	LOW
• REST API and webhooks for power users and integrations	LOW

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SUCCESS METRICS & KPIS

North Star, Product, Business

Product Health Metrics

Metric	Target / Definition	Type
Daily Active Operators (DAO)	70% of paid users log in every day	NORTH STAR
Payments Recorded / Day	Average 8 or more payments per paid operator per day	CORE
WhatsApp Receipt Delivery Rate	More than 95% delivered within 30 seconds of payment	QUALITY
Overdue Detection Time	Admin sees newly overdue student within 24 hours of due date	QUALITY
PDF Receipt Generation Time	Under 3 seconds at p95	PERFORMANCE
Parent App DAU/MAU	Above 40% — sticky daily parent engagement	ENGAGEMENT
AI Feature Adoption	More than 30% of Growth tier users use Smart Summary weekly	ADOPTION
Onboarding Completion	More than 75% of signups add first student and first payment	ACQUISITION
Support Ticket Rate	Fewer than 2 tickets per 100 active users per month	QUALITY

Business Metrics

Metric	Target
MRR Growth	20% month-over-month for the first 12 months
Free → Paid Conversion	More than 20% of free tier users convert to paid within 30 days

Metric	Target
Monthly Churn (paid)	Below 4% — solo operators are highly sticky once routine is established
ARPU	■350 per month blended average by Month 6
NPS	Above 50 by 6 months post-launch
CAC	Below ■500 per paid operator via organic WhatsApp referral and Play Store SEO
LTV:CAC Ratio	Above 7x by end of Year 1
Referral Rate	More than 25% of new signups arrive through an existing user referral

Full power. Single admin. Zero compromise.